

Groups and Lie Algebras: Exercise Sheet 6

1. Examples of Lie groups

Show that $GL(n, \mathbb{R})$, $O(n)$, $SO(n)$, $U(n)$, and $SU(N)$ are Lie groups. For this, you can use the implicit function theorem, which states that the preimage $F^{-1}(n)$ of a smooth map $F: M \rightarrow N$ at a point $n \in N$ is a manifold if for each $m \in F^{-1}(n)$ the differential $(DF)|_m: T_m M \rightarrow T_n N$ is surjective.

2. Tangent spaces at the identity

The Lie groups in the previous examples are all submanifolds of a vector space of matrices. The tangent space to $F^{-1}(n)$ at a point m can be identified with the kernel of the map $DF|_m: T_m M \rightarrow T_n N$. Compute the tangent space to the identity in (some of) the above examples.

3. $SO(3)$ representations, spherical harmonics, and the hydrogen atom

The spherical harmonics are given by

$$Y_l^m(\theta, \phi) = e^{im\phi} P_l(\cos \theta) N$$

where $P_l(\cos \theta)$ is a Legendre polynomial and $m \in \{-l, \dots, l\}$. For fixed l the spherical harmonics span the subspace $\mathcal{H}^{(l)}$ of $L^2(S^2)$. The spherical coordinates are picked as usual.

- Consider a rotation $(\theta, \phi) \mapsto R(\theta, \phi)$, $R \in SO(3)$. Then

$$Y_l^m(\theta, \phi) = \sum_{m'=-l}^l D_{mm'}^{(l)}(R) Y_l^{m'}(R(\theta, \phi)),$$

where $D_{mm'}^{(l)}(R)$ specify $(2l+1) \times (2l+1)$ dimensional matrices. For fixed l , what is the form of $D_{mm'}^{(l)}(R)$ for a rotation by an angle α around the z axis?

- This shows that $\mathcal{H}^{(l)}$ is a representation of $SO(3)$. Is it irreducible?
- Compute the action of the Hamiltonian for the hydrogen atom on $L^2(\mathbb{R}) \otimes \mathcal{H}^{(l)}$. (We have seen in the lecture that it commutes with the action of $SO(3)$ and hence preserves these subspaces).